



Sparse Matrices

SPARSE MATRICES

Represent two sparse matrices (mostly zeros) using arrays of triples [row, col, value]. Read dimensions and non-zero terms for two matrices. Add them, store result in sparse form (sorted by row then col), and display dense matrix form. Max 10x10; ignore zeros.

Sample Input (Matrix 1):

```
3 3 3
0 1 5
1 0 8
2 2 2
```

Sample Input (Matrix 2):

```
3 3 2
0 1 3
2 2 7
```

Sample Output (Sum Dense):

```
0 8 0
8 0 0
0 0 9
```

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 struct Triple {
4     int r, c, v;
5 };
6
7 int main() {
8     struct Triple a[50], b[50], s[100];
9     int m, n, t1, t2, i, j, k = 0;
10    int d[10][10] = {0};
11
12    scanf("%d%d", &m, &n, &t1);
13
14    for (i = 0; i < t1; i++)
15        scanf("%d%d", &a[i].r, &a[i].c, &a[i].v);
16
17    scanf("%d%d", &m, &n, &t2);
18
19    for (i = 0; i < t2; i++)
20        scanf("%d%d", &b[i].r, &b[i].c, &b[i].v);
21
```

OUTPUT

```
0 8 0
8 0 0
0 0 9
```

INPUT

```
3 3 3
0 1 5
1 0 8
2 2 2
```